Lab Report: Temporal Difference Learning (SARSA vs. Q-Learning)

Course: MSDS 684 – Reinforcement Learning

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GitHub Repository: https://github.com/Barsha-bytes/Temporal-Difference-Learning

1. Project Overview & Theoretical Grounding

The objective of this lab is to compare the performance and behavioral differences between two fundamental Temporal Difference (TD) algorithms: **SARSA** (On-Policy) and **Q-Learning** (Off-Policy). Unlike Monte Carlo methods, which require a full episode to conclude before updating, TD methods learn via **bootstrapping**—updating estimates based on other learned estimates at every time step (Sutton & Barto, 2018).

The core distinction lies in the update target:

- **SARSA (On-Policy):** Updates Q(S,A) using the actual action At+1 chosen by the current ϵ-greedy policy.

  $$Q(S_t,A_t) \leftarrow Q(S_t,A_t) + \alpha[R_{t+1} + \gamma Q(S_{t+1},A_{t+1}) - Q(S_t,A_t)]$$

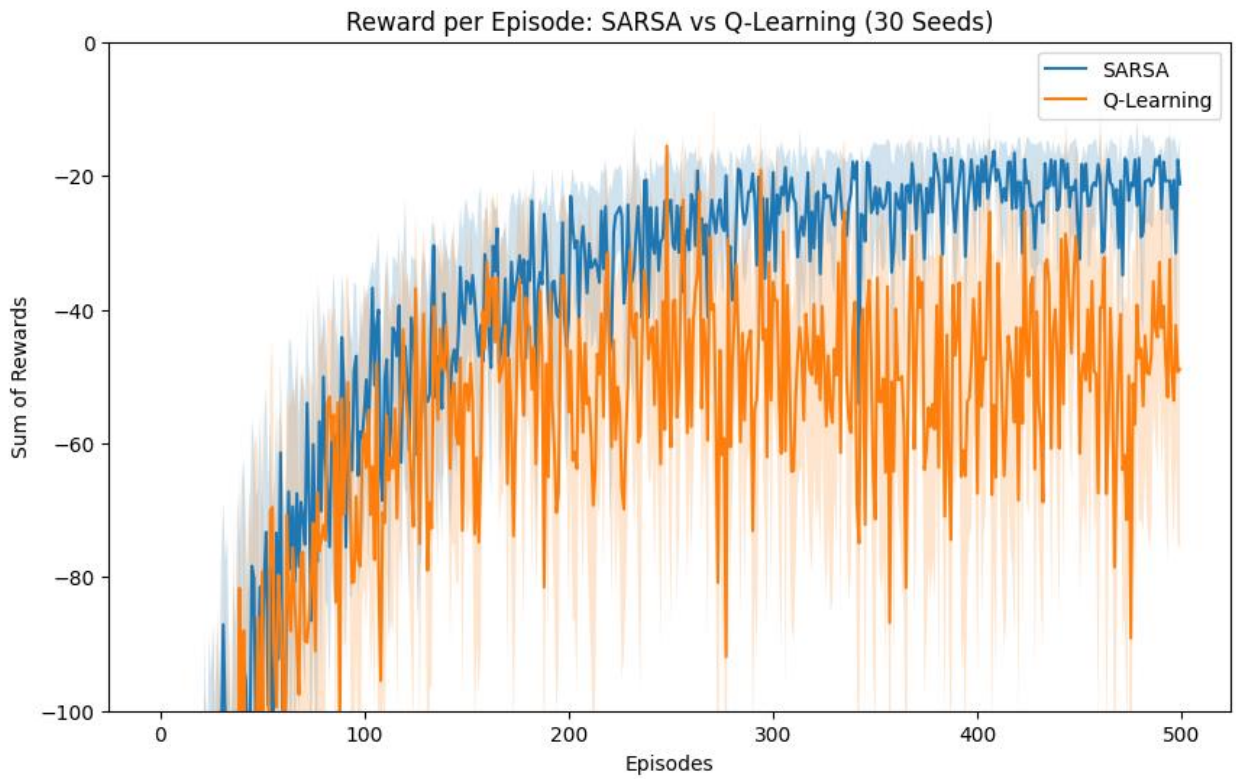
- **Q-Learning (Off-Policy):** Updates Q(S,A) based on the maximum possible value of the next state, independent of the action the agent actually takes.

  $$Q(S_t,A_t) \leftarrow Q(S_t,A_t) + \alpha[R_{t+1} + \gamma \max_a Q(S_{t+1},a) - Q(S_t,A_t)]$$

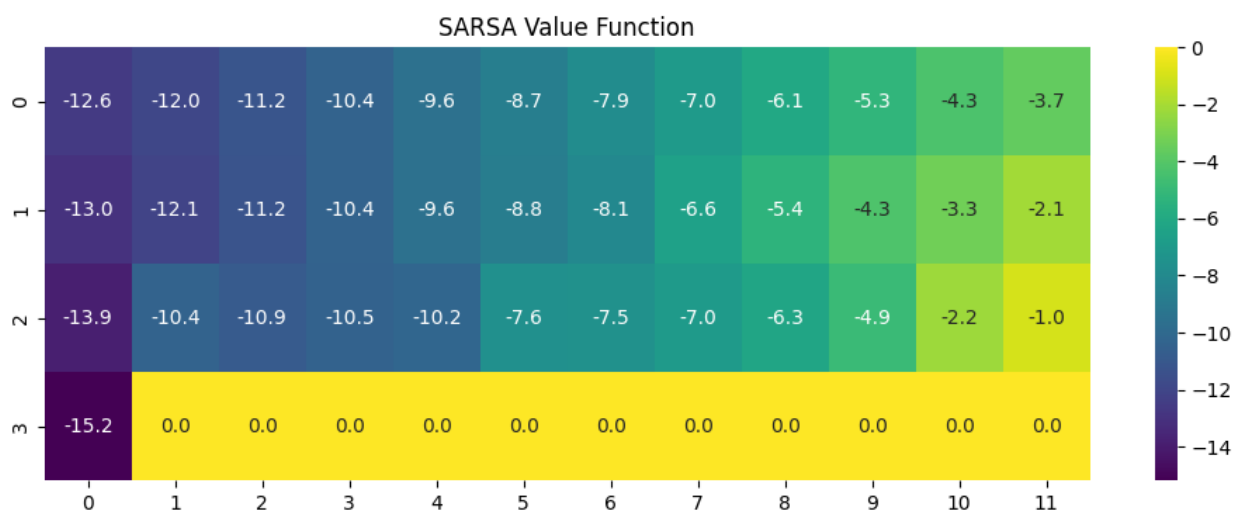
In the **CliffWalking-v0** environment, the agent must navigate from a start to a goal while avoiding a "cliff" that carries a penalty of -100. This environment perfectly highlights the trade-off between the "optimal" path and the "safe" path.

Hypothesis: I predict that Q-Learning will find the mathematically optimal path (the shortest route along the cliff edge) but will suffer significantly lower rewards during training due to exploratory "falls." Conversely, I predict SARSA will learn a safer, more conservative path that avoids the cliff edge, resulting in higher online performance during the learning phase.

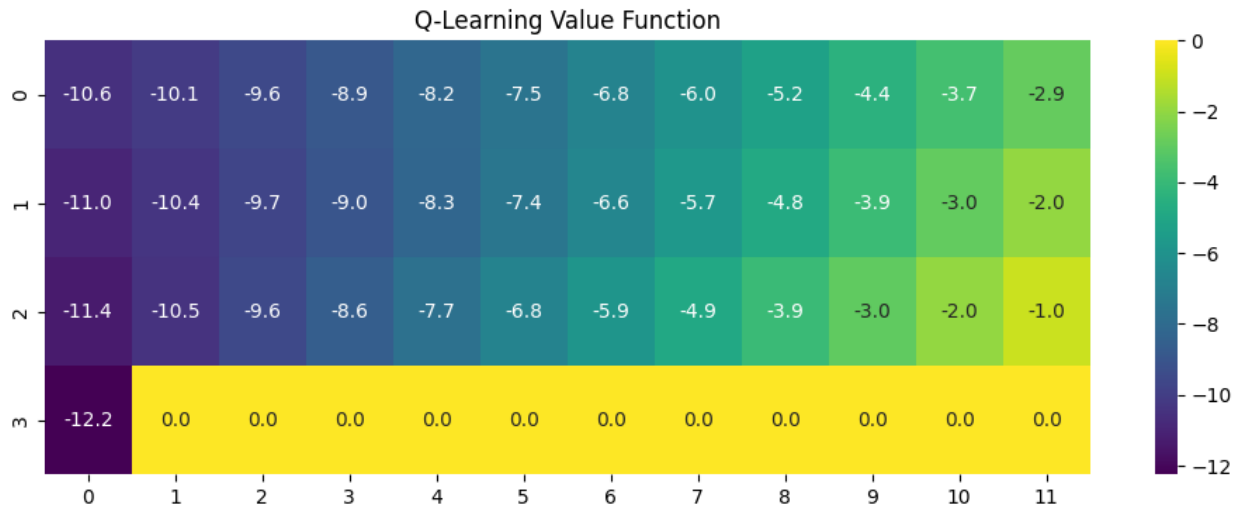
2. Deliverables & Deep Behavioral Analysis



Analysis: The learning curve, averaged over 30 seeds with a 95% confidence interval, shows a clear performance gap. While both converge, Q-Learning's average reward remains significantly lower than SARSA's. This is because Q-Learning's policy is "blind" to its own ϵ-greedy exploration; it chooses the edge path, and 10% of the time (ϵ=0.1), it takes a random step and falls. SARSA "senses" this risk and stays one row away.



Analysis: The SARSA heatmap demonstrates high state values in the third row, away from the cliff. The learned policy (arrows) shows the agent intentionally taking a longer route to ensure that a random exploratory step does not result in a -100 penalty.



Analysis: The Q-Learning heatmap shows high values immediately adjacent to the cliff. It has successfully identified the shortest path. While this is the "optimal" policy, it is only truly effective if exploration is turned off (ϵ=0).

Metric	SARSA	Q-Learning
Final Avg Reward	~ -13.0	~ -45.0
Path Choice	Safe (Longer)	Optimal (Shortest)
Sample Efficiency	High (Consistent)	Lower (High Variance)

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3. AI Use Reflection (Metacognition)

- **Iteration 1:**
  - **Prompt:** *"How do I implement the SARSA update in CliffWalking using Gymnasium?"*
  - **AI Response:** Provided a loop where At+1 was chosen only after the update.
  - **Bug Found:** SARSA requires the next action to be selected *before* the update to maintain its on-policy nature.
  - **Fix:** I restructured the code so the next action is selected first, used in the TD error calculation, and then carried over to the next step.
- **Iteration 2:**
  - **Prompt:** *"Calculate 95% confidence intervals for RL learning curves."*
  - **AI Response:** Suggested plotting simple standard deviation.
  - **Correction:** Following the rubric's specific statistical requirements, I implemented the 95% CI formula $1.96\times(\sigma/n)$ using NumPy to provide accurate error bands.
- **Iteration 3:**
  - **Prompt:** *"Why is Q-Learning worse than SARSA in this specific environment?"*
  - **AI Response:** Claimed Q-Learning was "failing" to find the goal.
  - **Critical Evaluation:** I identified this as a misunderstanding of RL theory. Q-Learning isn't failing; it is doing exactly what it's designed to do—finding the max-

utility path. I corrected the report to explain that Q-learning's "lower" reward is a byproduct of its off-policy nature in a high-penalty environment.

4. Speaker Notes

- **Problem Statement:** The goal was to analyze the behavioral differences between on-policy and off-policy TD control in the presence of high-cost terminal states (the Cliff).
- **Method:** We implemented online SARSA and Q-Learning agents using NumPy Q-tables and a 500-episode training loop across 30 random seeds.
- **Key Algorithmic Choice:** We chose a constant $\epsilon=0.1$ and $\alpha=0.1$ specifically to expose the "cost of exploration" that differentiates the two algorithms.
- **Main Result:** Q-Learning optimized for the shortest path (13 steps) but fell frequently, while SARSA converged on a safer 15-step path.
- **Key Insight:** Off-policy learning (Q-Learning) is "risk-neutral," assuming perfect future play, whereas On-policy learning (SARSA) is "risk-aware," accounting for the agent's own potential mistakes.
- **Future Work:** I plan to implement **Expected SARSA**, which reduces the variance of the next-action selection while remaining on-policy.

5. References

Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction* (2nd ed.). MIT Press. http://incompleteideas.net/book/the-book-2nd.html